MILITARY SPECIFICATION SHEET

VOLTMETER, DC, AIRCRAFT, DIGITAL DISPLAY INTEGRAL LIGHTED, 38 10 MILLIMETERS

This specification sheet is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the voltmeter described herein shall consist of this specification and the latest issue of MIL-V-85603

Requirements

Military part number Military part numbers of the voltmeters shall be as follows

Military Part Number	Type of Voltmeter
D85603/1-1	White integrally lighted (5VAC 400Hz)
D85603/1-2	Red integrally lighted (28VDC)
D85603/1-3	"NVIS Green A" integrally lighted (5VAC 400Hz)
D85603/1-4	"NVIS Green A" integrally lighted (28VDC)

Connector The minimum requirements for the voltmeter electrical receptacle shall be in accordance with MIL-C-38999, Series III, Class W, Finish Y, quick coupling, and Figure 1 herein

Integral lighting The voltmeter (D85603/1-1) shall be integrally lighted Instrument and Panel Lighting (IPL) white, the color shall be in accordance with MIL-P-7788 The voltmeter (D85603/1-2) shall be integrally lighted red in accordance with MIL-L-25467 The lighting circuitry and brightness shall be in accordance with the requirements of MIL-L-25467 Voltmeter (D85603/1-3) and voltmeter (D85603/1-4) shall be Night Vision Imaging System (NVIS) compatible in accordance with the requirements of MIL-L-85762 The contrast ratio of voltmeters shall not be less than 5 over the range of 1 0 foot-candle to 10,000 foot-candles.

Low voltage warning The low voltage warning threshold voltage is 21 0 volts

AMSC N/A FSC 6625

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Dimensions The voltmeter shall conform to the dimensional and display requirements of Figure 1 herein

Weight. The weight of the voltmeter when fully assembled and configured for in-service use shall be not greater than 0.23 kilogram

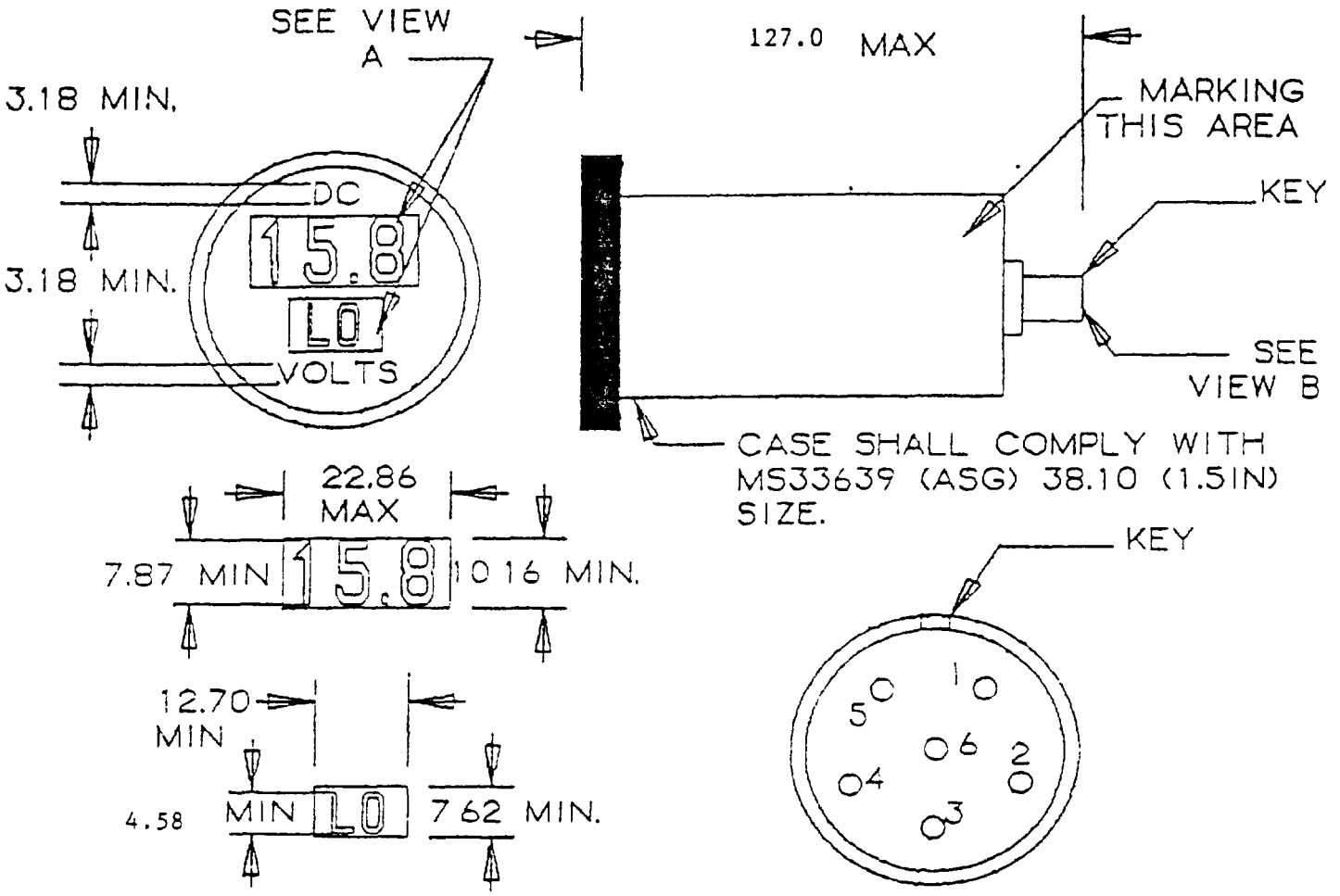
Electromagnetic interference The voltmeter shall be tested in accordance with MIL-STD-462 for those tests required by MIL-STD-461 for Class A1, Category A1b equipment The failure of the voltmeter to meet any of the requirements of the following tests shall be cause for rejection CE01, CE03, CS01, CS02, CS06, RE01, RE02, RS01, RS02, and RS03 except that the test levels for RS03 shall be as follows

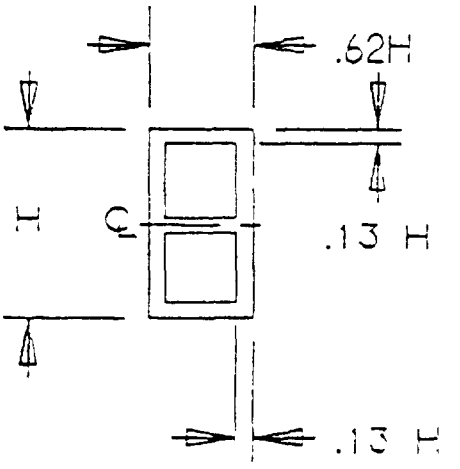
E-Field	Frequency Range
20V/meter	14kHz to 200MHz
65V/meter	200MHz to 450MHz
20V/meter	450MHz to 1.0GHz
100V/meter	1.0GHz to 12 0GHz
20V/me-	12 0GHz to 40 0GHz except 16 5GHz and 33 0GHz
100V/meter	16 5GHz and 33 0GHz

Integral lighting The voltmeter (D85603/1-1) shall be subjected to and pass the tests of MIL-P-7788 and voltmeter (D85603/1-2) shall be subjected to and pass the tests of MIL-L-25467 Both voltmeters shall comply to the requirements of MIL-L-25467 for brightness (illumination), stray light, and visibility of displays Voltmeter (D85603/1-3) and voltmeter (D85603/1-4) shall be tested in accordance with MIL-L-85762 to insure compatibility with Night Vision Imaging Systems (NVIS) The voltmeters shall not exhibit a contrast ratio of less than 5 over the range of 1 0 foot-candle to 10,000 foot-candles During these tests, the voltmeter shall be energized and the input voltage adjusted to 15 8 VDC When required, the lighting circuit shall be energized with the appropriate voltage

Preparing Activity
Navy - AS
(Project No 6625- N973-01)

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D38999/21YA35PN
PIN DESIGNATIONS

1- INPUT(+)AIRCRAFT BATTERY
2- INPUT(-)AIRCRAFT BATTERY
3- CAUTION ADVISORY SWITCH (ACFT PANEL)
4- CAUTION ADVISORY SWITCH (GND)
5- LIGHTING INPUT (HIGH)
6- LIGHTING INPUT (LOW)

DIMENSIONS IN MILLIMETERS. TOLERANCE UNLESS OTHERWISE SPECIFIED: XX± .76

FIGURE 1- Aircraft digital voltmeter.

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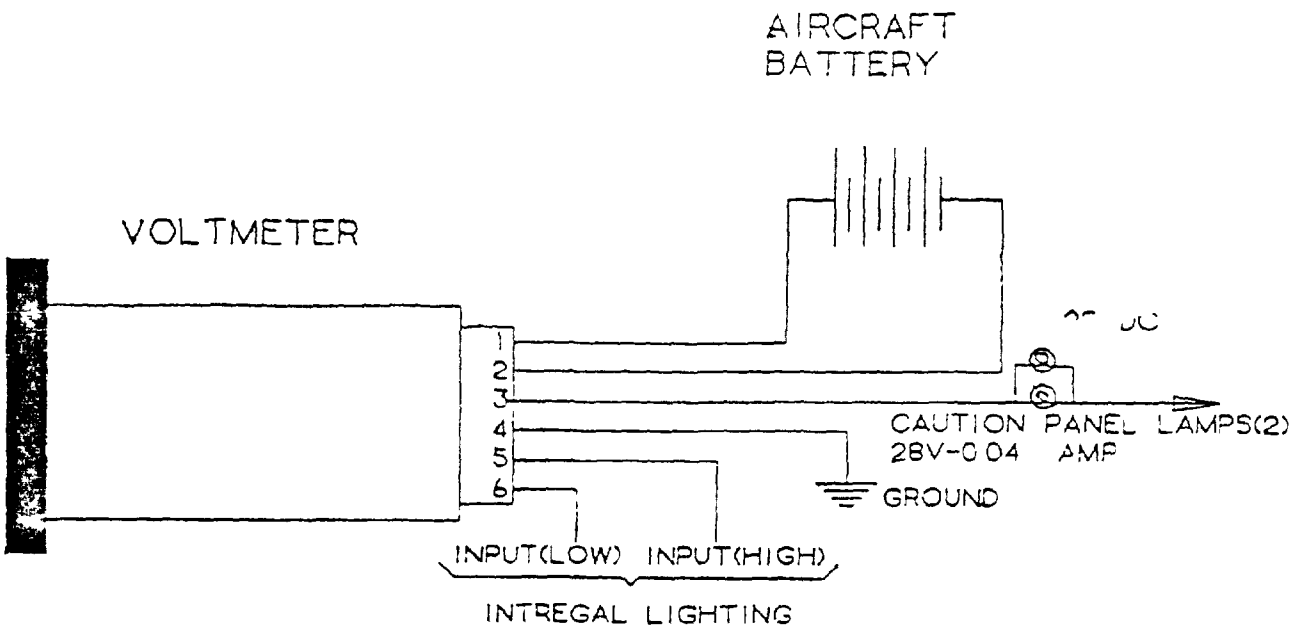


FIGURE 2- External circuitry.

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